

VLAN Configuration and Router-on-a-Stick Lab Notes

1. Topic Studied

Inter-VLAN Routing using VLANs, Trunk Links, and Router-on-a-Stick Configuration

The goal of the lab was to separate departments into different VLANs and allow communication between them using a router.

2. VLAN Overview

A **VLAN (Virtual Local Area Network)** divides a physical switch into multiple logical networks.

Benefits:

- Improves security
- Reduces broadcast traffic
- Separates departments
- Makes network management easier

Example VLAN structure:

VLAN ID	Department
10	TSG
20	ACCOUNTING
30	PST
40	MIS
50	SORTING
60	PRINTING

3. Creating VLANs on a Switch

Example:

```
enable
configure terminal
```

```
vlan 20
name ACCOUNTING
```

```
vlan 50
name SORTING
```

```
vlan 60
name PRINTING
```

```
end
```

Verify VLANs:

```
show vlan brief
```

4. Assigning Switch Ports to VLANs

Access ports connect end devices such as PCs.

Example on Switch03:

configure terminal

```
interface fa0/1
switchport mode access
switchport access vlan 60
```

```
interface fa0/2
switchport mode access
switchport access vlan 50
```

```
interface fa0/3
switchport mode access
switchport access vlan 20
```

end

Verification:

```
show vlan brief
```

Expected result:

VLAN 20	ACCOUNTING	Fa0/3
VLAN 50	SORTING	Fa0/2
VLAN 60	PRINTING	Fa0/1

5. Trunk Configuration

A trunk carries multiple VLANs between switches or between a switch and router.

Example:

```
interface fa0/4
switchport mode trunk
```

Check trunk status:

```
show interfaces trunk
```

A working trunk should show:

```
Status: trunking
Allowed VLANs: 1,10,20,30,40,50,60
```

6. Router-on-a-Stick Configuration

A router uses subinterfaces to provide gateways for each VLAN.

The physical interface:

GigabitEthernet0/0/0

Subinterfaces created:

VLAN	Gateway
10	192.168.10.1
20	192.168.20.1
30	192.168.30.1
40	192.168.40.1
50	192.168.50.1
60	192.168.60.1

Example:

```
interface gi0/0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
```

Repeat for other VLANs.

Verification:

```
show ip interface brief
```

Successful output:

```
Gi0/0/0.10 192.168.10.1 up/up
Gi0/0/0.20 192.168.20.1 up/up
Gi0/0/0.30 192.168.30.1 up/up
Gi0/0/0.40 192.168.40.1 up/up
Gi0/0/0.50 192.168.50.1 up/up
Gi0/0/0.60 192.168.60.1 up/up
```

7. PC IP Configuration

Each PC needs:

Accounting PC (VLAN 20)

IP Address: 192.168.20.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.20.1

Sorting PC (VLAN 50)

IP Address: 192.168.50.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.50.1

Printing PC (VLAN 60)

IP Address: 192.168.60.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.60.1

8. Troubleshooting Commands

Check VLANs

```
show vlan brief
```

Check trunks

```
show interfaces trunk
```

Check router interfaces

```
show ip interface brief
```

Test connectivity

From a PC:

```
ping 192.168.x.1
```

Example:

```
ping 192.168.20.1
```

9. Problems Found During the Lab

Problem 1:

Configured:

```
interface fa0/0.10
```

Error:

Invalid interface type and number

Cause:

The router did not have a FastEthernet 0/0 interface.

Solution:

Use the correct interface:

```
interface gi0/0/0.10
```

Problem 2:

Switch03 VLANs existed but no ports were assigned.

Before:

```
20 ACCOUNTING active
50 SORTING active
60 PRINTING active
```

No interfaces were shown.

Solution:

Assign PC ports:

```
switchport access vlan <number>
```

10. Final Network Checklist

- ✓ VLANs created
- ✓ PC ports assigned to VLANs
- ✓ Switch-to-switch links configured as trunks
- ✓ Switch-to-router link configured as trunk
- ✓ Router subinterfaces created
- ✓ PC IP addresses configured
- ✓ Default gateways configured
- ✓ Ping tests successful

Main concept learned:

VLANs separate networks, trunks carry multiple VLANs, and router subinterfaces allow communication between different VLANs.